P1

```cpp
#include <bits/stdc++.h>
using namespace std;

long long factorial(int n) {
    long long res = 1;
    for (int i = 1; i <= n; i++) {
        res *= i;
    }
    return res;
}

int main() {
    int n, k;
    cout << "Enter a positive integer for n (1 to 20): ";
    cin >> n;
    cout << "Enter a positive integer for k (k<=n); ";
    cin >> k;

    long long p = factorial(n) / factorial(n - k);
    cout << "Permutation of n object k at a time: " << p << endl;

    return 0;
}
```

```
PS C:\Users\Leo\Desktop\coding\Problems\yzu\CS106\HW_5> .\P1.exe
Enter a positive integer for n (1 to 20): 6
Enter a positive integer for k (k<=n); 4
Permutation of n object k at a time: 360
```

P2

```cpp
#include <bits/stdc++.h>
using namespace std;

int pascal(int row, int col) {
    if (col == 0 || col == row) {
        return 1;
    }
    return pascal(row - 1, col - 1) + pascal(row - 1, col);
}

int main() {
    int n;
    cout << "Enter the size of the Pascal triangle: ";
    cin >> n;
    cout << "The coefficients are shown below:" << endl;

    for (int i = 0; i <= n; i++) {
        for (int j = 0; j <= i; j++) {
            cout << pascal(i, j) << " ";
        }
        cout << endl;
    }

    return 0;
}
```

```
PS C:\Users\Leo\Desktop\coding\Problems\yzu\CS106\HW_5> .\P2.exe
Enter the size of the Pascal triangle: 5
The coefficients are shown below:
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
```

P3

```cpp
#include <bits/stdc++.h>
using namespace std;

bool isPrime(int n) {
    if (n < 2) return false;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) return false;
    }
    return true;
}

int reverseNum(int n) {
    int rev = 0;
    while (n > 0) {
        rev = rev * 10 + n % 10;
        n /= 10;
    }
    return rev;
}

bool isEmirp(int n) {
    return isPrime(n) && isPrime(reverseNum(n));
}

void printEmirp() {
    cout << "The emirp numbers less than 1000:" << endl;
    int count = 0;
    for (int i = 2; i < 1000; i++) {
        if (isEmirp(i)) {
            cout << i << " ";
            count++;
            if (count % 10 == 0) {
                cout << endl;
            }
        }
    }
    if (count % 10 != 0) {
        cout << endl;
    }
}

int main() {
    printEmirp();
    return 0;
}
```

```
PS C:\Users\Leo\Desktop\coding\Problems\yzu\CS106\HW_5> .\P3.exe
The emirp numbers less than 1000:
2 3 5 7 11 13 17 31 37 71
73 79 97 101 107 113 131 149 151 157
167 179 181 191 199 311 313 337 347 353
359 373 383 389 701 709 727 733 739 743
751 757 761 769 787 797 907 919 929 937
941 953 967 971 983 991
```

P4

```cpp
#include <bits/stdc++.h>
using namespace std;

bool isDivisible(int num, int divisor) {
    return num % divisor == 0;
}

int main() {
    int n;
    cout << "Enter a number between 1 and 100: ";
    cin >> n;

    cout << "The number: " << n << endl;
    cout << "The factors: ";
    for (int i = 1; i <= n; i++) {
        if (isDivisible(n, i)) {
            cout << i << " ";
        }
    }
    cout << endl;

    return 0;
}
```

```
PS C:\Users\Leo\Desktop\coding\Problems\yzu\CS106\HW_5> .\P4.exe
Enter a number between 1 and 100: 60
The number: 60
The factors: 1 2 3 4 5 6 10 12 15 20 30 60
```